

A simulation study to investigate the effect of variation of blood-suspending medium on the photoacoustic signal

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Abstract:

The study of blood has been a subject of great interest for decades in understanding human physiology. The optical properties [absorption coefficient (μ_a), scattering coefficient (μ_s), and anisotropy coefficient (g)] of blood play key role in generation and determining the strengths of photoacoustic (PA) waves. Red blood cells (RBCs) are the dominating PA sources in blood and the magnitudes of these optical parameters depend on the medium in which RBCs are suspended. In this simulation study, two types of media [phosphate buffer saline (PBS) and blood plasma (PLS)] have been investigated herein for excitation wavelengths involving 480, 650, and 1050 nm and for five different hematocrit levels, Hct=10–45%. The distinguishability of the PA response lines for PBS and PLS media is much more prominent ($\approx 17\%$ at Hct=45%) at 650 nm than those of 480 and 1050 nm. The reason might be the variation of g factor ($\approx 0.8\%$) between two types of samples at 650 nm, whereas g almost remains the same at other wavelengths. Our computational study might be helpful in understanding the effect of the blood-suspending medium on PA signal properties at multiple optical wavelengths.

Keywords:

Blood plasma (PLS), phosphate puffer saline (PBS), photoacoustics (PAs), red blood cells (RBCs), absorption coefficient (μ_a), scattering coefficient (μ_s), anisotropy coefficient (g), Monte Carlo (MC) method.

Introduction:

Blood is extremely important for a living human body which contains cellular components, including red blood cells (RBCs), white blood cells (WBCs), platelets, and blood plasma as the suspending medium [1]. RBCs are responsible for transporting O_2 , CO_2 and nutrients to keep the body cells alive, WBCs are important to develop immunity, and platelets stop bleeding from any cut in our body. Plasma (PLS) helps cells to move smoothly from one part to other parts of our body. It is yellowish in color, contains several vital proteins, vitamins, minerals, ions (e.g., Ca^{+2} , Mg^{+2} , K^{+}) etc. These proteins contribute $\approx 10\%$ of its volume and 90% of its content is water [2]. There are significant person to person variations in optical properties of whole blood and PLS. The phosphate puffer saline (PBS) medium, most commonly used medium for resuspending the washed cells, is expected to exhibit different optical properties than that of the PLS medium because of the presence of solid complex macromolecules. The suspending medium (PBS/PLS) affects the bulk optical properties, namely absorption coefficient (μ_a), scattering coefficient (μ_s), and anisotropy coefficient (g) of blood [3, 4, 5]. Researchers all over the world have been employing extensively the photoacoustic (PA) technique to study different aspects of blood [6-9]. We have observed that the roles of μ_s and g

on PA emission from blood have not been examined in details. Therefore, we have conducted a simulation study to investigate the effect of the suspending medium on PA emission from blood at 480, 650 and 1050 nm optical wavelengths.

Simulation framework:

The simulation study was conducted into two parts- (i) Monte Carlo simulation followed by (ii) k-Wave simulation.

(A) **Monte Carlo simulation:** The Monte Carlo model (MC) works based on the Radiative Transfer Equation (RTE) to understand the propagation of photons through a biological tissue or blood sample [10, 11]. The RTE is hard to solve analytically except for very simple and symmetric shapes like sphere, cylinder, and ellipsoid. The MC simulations were carried out in this work to estimate the optical fluence maps in different blood samples.

(B) **k-Wave simulation:** The k-Wave simulation was performed to calculate the PA signal by solving equation (1) throughout the sample for a given fluence distribution. The time-dependent PA equation in 3D can be cast as,

$$\nabla^2 P(\mathbf{r}, t) - \frac{1}{v_s^2} \frac{\partial^2}{\partial t^2} P(\mathbf{r}, t) = -\frac{\beta}{C_p} \frac{\partial H(\mathbf{r}, t)}{\partial t}, \quad (1)$$

where β is the isobaric volume expansion coefficient, C_p is the isobaric specific heat, $H(\mathbf{r}, t)$ is the heating function, and $P(\mathbf{r}, t)$ is the pressure at an arbitrary point $\mathbf{r}$ and at time t . Then the peak-to-peak pressure can be determined as $P_{pp} = \max[P(t)] - \min[P(t)]$. In the k-Wave simulations, the surrounding medium was chosen as water.

(C) **Choice of parameters:** To start with the simulation work, we calculated the optical parameters of blood at different hematocrit (Hct) levels with fixed oxygen saturation level, $\text{SpO}_2=70\%$ (vein blood). The useful formulas to calculate μ_a , μ_s , and g are discussed in [6]. In MC simulation, the sample dimension was taken as, $(n_x, n_y, n_z) = (100, 100, 100)$. The individual voxel volume was $dx \times dy \times dz = (0.005)^3 \text{ cm}^3$. The diameter of the incident photon beam was, $D_B = 0.05 \text{ cm}$ and the number of photons was fixed to 3 million. The simulations were executed in a PC with specifications: Intel(R) Core(TM) i5-9500, 3.00 GHz, 6 cores, 6 processors, Microsoft Windows 11 Pro OS, 16.0 GB RAM (GCC-6.3.0-1 and MATLAB 2022b). The k-Wave computational domain had dimensions of $(N_x, N_y, N_z) = (120, 120, 170)$ grid points and the voxel volume was $dx \times dy \times dz = (0.005)^3 \text{ cm}^3$. The numerical values of the thermodynamical parameters (C_p , β) were set to 1. The speed of sound and density of the sample and surrounding medium were taken as, $v_s = v_m = 1500 \text{ m/s}$ and $\rho_s = \rho_m = 1000 \text{ kg/m}^3$, respectively. The thickness of the perfectly matched layer (PML) was chosen to be 0.05 cm. An attenuation coefficient of 100 np/m inside the PML layer was considered. A circular aperture ultrasound transducer of radius 0.05 cm was placed at the coordinates $(N_x/2, N_y/2, N_z-20)$. A total 31 voxels essentially built the sensor. The center frequency and fractional bandwidth of the sensor points were set to be 7.5 MHz and 70%, respectively.

Simulation results:

It is seen from Fig. 1 (A-C) that the strength of μ_a is the highest at 480 nm at each Hct level among the all wavelengths. Figure 1 (D-F) illustrates that μ_s at 650 nm is slightly higher compared to those of 480 and 1050 nm (lowest). The separation between the μ_s curves is less at 650 nm compared to those of the other wavelengths. Further, g value for RBC+PLS is 0.8% greater than that of RBC+PBS at 650 nm, whereas g lines are non-separable at 480 and 1050 nm. The PA signals for a particular hematocrit level (Hct=30%) for three wavelengths are shown in Fig. 2. RBC+PBS and RBC+PLS signals seem to be overlapped at 480 and 1050 nm but a slight gap exists at 650

nm. The PA amplitude decreases as the wavelength increases (i.e., 480, 650, and 1050 nm). In Fig. 3, P_{pp} versus Hct plot shows that the difference between RBC+PBS and RBC+PLS lines at 650 nm is visibly prominent compared to those of 480 and 1050 nm.

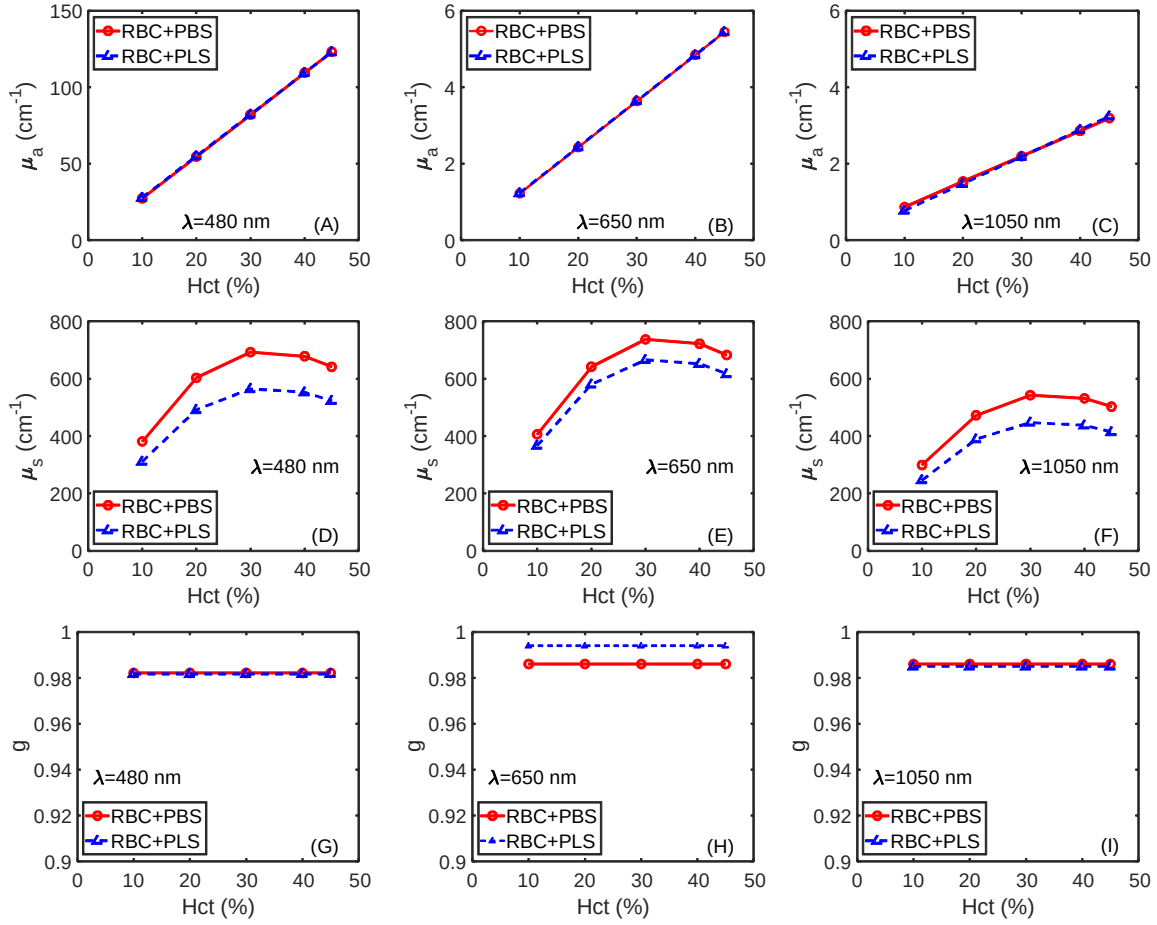


FIGURE 1: (A)-(C) The plots of μ_a with Hct. (D)-(F) The variation of μ_s as a function of Hct. (G)-(I) Same as (D)-(F), respectively but for the anisotropy coefficient g .

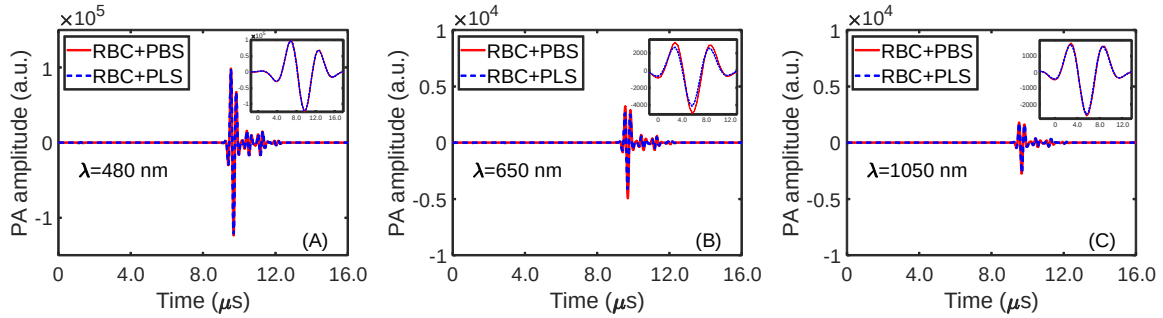


FIGURE 2: (A)-(C) The PA signals at Hct=30% for the PBS and PLS suspending media at 480, 650, and 1050 nm wavelengths, respectively.

Discussion and conclusions:

In this simulation work, our motivation is to investigate the effect of the blood-suspending medium on PA signal. Optical parameters (μ_a , μ_s and g) for a series of blood samples have been theoretically estimated. These parameters

are utilized to obtain the photon depositions in numerical samples which have been subsequently used to execute the k-Wave simulations to compute the PA signals. Figure 1(A-C) shows that μ_a is the highest at 480 nm, followed by 650 nm and lowest at 1050 nm for both the suspending media. The absorption coefficients of PBS and PLS media are very low ($\approx 10^3$ order of magnitude less) compared to RBCs [4]. Note that μ_a values for both the PBS and PLS media at 650 nm are negligibly small; μ_a of PLS at 480 nm is higher than that of PBS; however, the opposite trend is observed at 1050 nm [4].

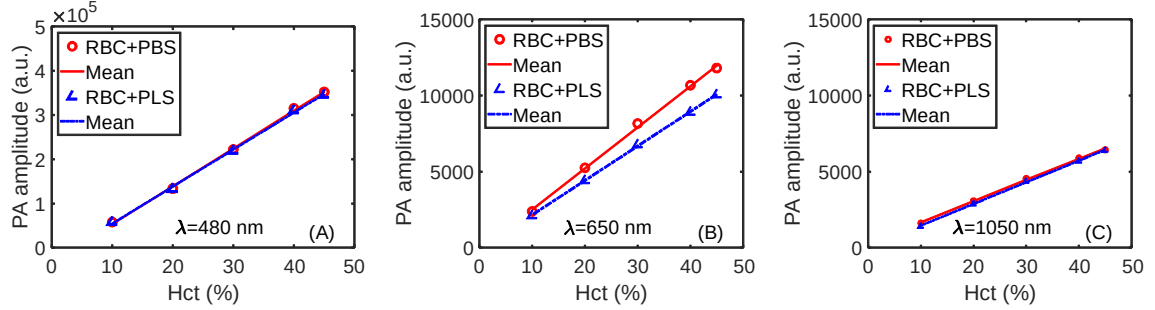


FIGURE 3: (A)-(C) The plots of PA peak-to-peak values at different Hct levels for the PBS and PLS suspending media at 480, 650, and 1050 nm wavelengths, respectively.

The PA lines for RBC+PBS and RBC+PLS samples at 650 nm are well separated [see Fig. 3(B)] though it is not observed at 480 and 1050 nm. For example, the amplitude changes by $\approx 18\%$ at Hct=47%. The most obvious reason behind this is the change of g-value at 650 nm (the changes in μ_a and μ_s are almost comparable at all three wavelengths). Therefore, a small change in g-value can induce a significant change in the PA response and this is the most important finding of this work. In other words, the effect of the surrounding medium becomes prominent at 650 nm. This simulation study reveals that g is a very sensitive optical parameter that greatly impacts the PA response of blood samples at certain wavelengths.

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